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Project Title:	In Vivo Induction Of Chromosomal Damage: A Spectral Karyotyping Study		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
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Zip Code:	11794-8691	**Congressional District:**	1
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
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COI Name (Institution):	Whorton, Elbert (UTMB at Galveston) Golightly, Marc (Stony Brook University)		
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Task Description:

Manned deep space missions that require prolonged flights may create potential health risks for late occurring disease (such as cancer) to the astronauts from exposure to various types of galactic cosmic rays such as high-energy and heavy particles (HZE). Among these, stripped 56Fe ions are of greatest concern because of their very high linear energy transfer (LET) characteristic resulting in wider cellular damage and the lack of information on in vivo biological effects of these particles. For these reasons, animal studies directed towards the understanding of cellular and molecular mechanisms potentially contributing to carcinogenesis of HZE particles are essential to improve the estimates of the health risk of protracted space flight. Moreover, for radiation-induced genetic damage studies to be useful for risk assessment, they must be established in in vivo systems especially in cells that are known to be at risk for health problems associated with radiation exposure such as bone marrow cells, the known target cell for radiation-induced leukemia.

This proposal is designed to characterize the type and distribution of initial and the subsequent loss of chromosomal damage, and the expression of genomic instability induced in vivo in bone marrow cells. This approach is the most relevant to assess health risks but has not been used with HZE particles. A spectral karyotyping (SKY) of mouse chromosomes will be used for the detection, analysis, and characterization of chromosome damage in bone marrow cells collected at different times (up to 3 months) post-irradiation. Two strains of mice known to be sensitive (CBA/CaJ) and resistant (C57BL/6) to radiation induced chromosome aberrations and cancer will be exposed to graded doses of HZE particles, i.e. 1 GeV/Amu 56Iron (Fe) ions or to low LET 137Cs gamma rays (as a source of reference radiation). Multiple sampling times permit an accurate determination of relative biological effectiveness (RBE) for 56Fe ion-induced chromosomal damage relative to 137Cs gamma rays, as a function of absorbed dose, aberration type, and time after irradiation. Flow cytometry will be used to evaluate the magnitude of cell cycle delay induced by 56Fe ions in the bone marrow cells. The result from flow cytometric analysis will provide the much-needed in vivo information for determining the influence of cell cycle arrest on the expression of chromosome damage in metaphase cells. The use of SKY for the analysis of all types of chromosomal damage allows a more accurate measurement of the frequency, degree of complexity, and a dose response for the production of simple and complex aberrations. This proposed in vivo approach is aimed to address the following hypotheses (1) Complex chromosomal exchanges are more prevalent in the bone marrow cells of mice exposed to 56Fe ions than those exposed to similar doses of 137Cs gamma rays. (2) There is a dose-dependent increase in the frequency of chromosome aberrations in the bone marrow cells of both strains of mice exposed to 56Fe ions or 137Cs gamma rays. (3) Genomic instability, expressed as delayed appearance of chromosome aberrations, and clonal expansion of chromosomal exchanges commonly found in mice exposed to 56Fe ions and 137Cs gamma rays are detected in a fraction of surviving cells at later times post-irradiation. (4) Genetic background influences the susceptibility to the induction of in vivo chromosomal damage and the subsequent expression of genomic instability. The result from this study will provide pivotal information that should lead to a better understanding of dose-response kinetics and of mechanisms by which 56Fe ions induced aberrations can potentially be involved in the induction of genomic instability and cancer. The result also would be important in improving risk assessment of protracted space flight.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

This project was initiated in late July 2003. The major goals during the first year are to: (1) conduct a preliminary study on flow cytometric analysis of cell cycle and apoptosis in bone marrow cells collected at different times up to 66 hours post-irradiation from mice exposed to different doses of HZE particles (56Fe ions of 1Gev/amu) or low LET ionizing radiation (137Cs gamma rays), (2) complete the exposure of mice to different doses of HZE particles or low LET ionizing radiation for collecting bone marrow cells at different time up to three months after irradiation for later analysis of chromosomal aberrations induced in vivo, and (3) initiate the analysis of chromosome aberrations in the bone marrow cells of mice collected at different times after irradiation, by means of the genome-wide multicolor fluorescence in situ hybridization of mouse chromosomes (mFISH) or spectral karyotyping (SKY).

We have completed the exposure of CBA/Ca and C57BL/6 mice to both types of radiation for cell cycle study and for the analysis of chromosome aberrations. In November 2003, we participated in the NSRL-1 run at BNL. Subsequently, we have completed gamma-irradiation of mice, using a GammaCell40 (located at Stony Brook University).

We also have completed the cell cycle analysis and presented the result at the 3rd International Workshop on Space Radiation Research and 15th Annual NASA Space Radiation Health Investigators' Workshop, May 16-20, 2004, Port Jefferson, New York. We also have submitted a manuscript based upon these data for possible publication in the Workshop Proceedings. Below is a brief detail of our results on in vivo effects of 56Fe ions on cell cycle.

We used flow cytometry to examine the magnitude of cell cycle delay and apoptosis in bone marrow (BM) cells collected at different times (18, 42 and 66 hours) from two strains of mouse (CBA/CaJ and C57BL/6J) following acute in vivo exposure to HZE particles (1 GeV/amu 56 Fe ions) or 137Cs gamma rays. At each sacrifice, BM cells were collected from three mice of each strain per dose of 56Fe ions (0, 10 and 100 cGy; at the dose rate of 1Gy/min) and two mice of each strain per dose of 137Cs gamma rays (0, 100 and 300 cGy; at the dose rate of 72 cGy/min). We found no effects on cell cycle by a single dose of 10 cGy 56Fe ions. However, we observed a significant decrease in the percentage (%) of cells cycling through S-phase after exposure to 100 cGy 56Fe ions in both CBA/CaJ ($P<0.005$) and C57BL/6J ($P<0.05$) mice relative to their corresponding sham controls at 18 hr post-irradiation. The effect was more pronounced in CBA/CaJ mice. The % of S-phase cells subsequently increased and persisted up to 66 hr post-irradiation. Similar results were found in mice exposed to 300 cGy 137Cs gamma rays but the magnitude of effect was less than that seen in mice exposed to 56Fe ions. With respect to apoptosis, higher numbers of apoptotic cells were observed in BM cells collected at 18 hr post-irradiation from both strains of mouse exposed to 56Fe ions relative to those collected from gamma-irradiated mice. At a similar dose level of 56Fe ions, we found a slightly higher % of early and late apoptotic cells in CBA/CaJ mice than that observed in C57BL/6J mice. The % of G2/M BM cells collected from mice exposed to either 100 cGy 56Fe ions or 300 137Cs gamma rays at 66 hr post-irradiation was consistently higher than that in their sham controls. The results provide useful information for future studies using metaphase analysis to examine in vivo induction of chromosomal damage following exposure to 56Fe ions. The results from this section of the study indicate differences in response to radiation between the CBA/Ca and C57BL/6 mice, suggesting the influence of genetic background on radiosensitivity. Therefore, the results address Critical Question #10.12 of NASA Critical Path Roadmap, i.e. what are the effects of age, gender and inter-individual diversity.

With respect to the analysis of chromosome aberrations, we have completed collecting bone marrow cells from both strains of mice exposed to either 56Fe ions or 137Cs gamma rays. Exposed mice were serially sacrificed at 2 days, 4

	days, 7 days, 1 month and 3 months following irradiation. At each sacrifice time, four groups of mice (five in each) in each strain were euthanized for bone marrow collection. Bone marrow cells were collected from each mouse, processed and stored in a form suitable for use in metaphase preparation that will span throughout this 3-year project. In early April 2004, we obtained a Zeiss microscope AxioPlan 2 equipped with a very sensitive cooled color charge-coupled device (CCD) camera for capturing multicolor-metaphase FISH images and delivery of the signals to a sophisticated digital image analysis computer software from MetaSystems Inc. (Belmont, MA) for the ultimate analysis of chromosome aberrations. At this stage, a period of training was given by the vendor to provide assurance that the most effective and reliable analysis of chromosomal aberrations is obtained from the use of this spectacular piece of equipment. We are now working diligently on the optimum use of this novel mouse mFISH digital analysis system. Due to a very large number of metaphase cells to be scored, the evaluation of genome-wide chromosome aberrations by mFISH will be continued into year 3 of the study. During the next years, studies will focus on the characterization of the type and distribution of initial and the subsequent loss of chromosomal damage, and the expression of genomic instability induced in vivo in bone marrow cells. Dose and time response relationships will be established for the induction of genomic instability.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Rithidech K, Golightly M. "Analysis of cell cycle in mouse bone marrow cells following acute in vivo exposure to 56Fe ions." The Proceedings of the 3rd International Workshop on Space Radiation Research, Port Jefferson, NY, May 16-20, 2004. Proceedings of the 3rd International Workshop on Space Radiation Research, Port Jefferson, NY, May 16-20, 2004. , Jun-2004
Presentation	Rithidech, K.; Golightly, M. "Analysis of cell cycle in mouse bone marrow cells following acute in vivo exposure to 56Fe ions: A Pilot Study" Poster Presentation, the 3rd International Workshop on Space Radiation Research and 15th Annual Space Radiation Health Investigators' Workshop, Port Jefferson, NY May-2004